2026 Science Symposium
Dickinson College

Nutrition and Diversity in Pennsylvania's Public-School Lunch Menus
Scraping menu items from Middle schools and High schools using Nutrislice

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Introduction

School provided meals account for nearly 50% of a child's daily caloric intake (Cullen & Chen, 2017), and lower income students are more reliant on school lunches than upper- and middle-income students (Cullen & Chen, 2017). Understanding how income and school lunch quality are correlated can help inform future policy and set goals to provide nutritious lunches regardless of socioeconomic status. This study examines the relationship between district-level income and the diversity and nutritional content of middle and high school lunches in Pennsylvania. Lunch menu data for March 2026 were collected from 91 middle and high schools using Nutrislice, a third-party menu platform that provides nutritional information for each item served. District-level income data were obtained from the Commonwealth of Pennsylvania Department of Education reports. Nutrition scores were calculated based on normalized nutritional content, and diversity scores were assigned based on the number of unique food items offered across entrée, vegetable, and fruit categories. Linear regression analysis revealed a weak but statistically significant positive relationship between district-level income and both diversity and nutrition scores. These findings suggest that while income contributes to variation in school lunch quality, it explains only a small portion of the observed differences, indicating that additional factors likely play a larger role.

Research Question

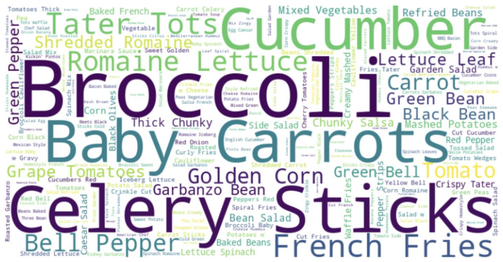
Does district-level income influence the diversity and nutritional content of middle and high school lunch menus in Pennsylvania?

What is Nutrislice?

Nutrislice is a digital platform and mobile app used by school districts to publish, manage, and display food service menus, nutritional data, and allergy information. It is a system used to share daily meal options with parents and students. Approximately 1/4 of PA school districts use Nutrislice (98).

Drawbacks of Nutrislice:

Nutrislice is a paid service for schools and costs can be significant depending on the size of the districts. The affordability of this platform is likely creating a bias in our data in which higher income districts are more likely to pay for this additional service to display their menus. The lack of consistent menu reporting across



Methods

Data Preparation

Webscraping Nutrislice
- Used selenium webdriver to search up school district on Nutrislice.
- Automated API link construction for each school
- Scraped the lunch menu of each school for the month of March

Data Cleaning
- Some schools were missing categories for their food items
- Only categorized foods in vegetable, fruit and entrée category were included in calculating diversity score and nutriscore.

Feature engineering
- Log transformed averaged income per district
- Calculated NutriScore and diversity scores per school

Data Merging
- Merged district level income with school level lunch menu nutrition and diversity data.

Linear Regression
- Conducted linear regression comparing the diversity and nutrition score to income.

Calculating NutriScore

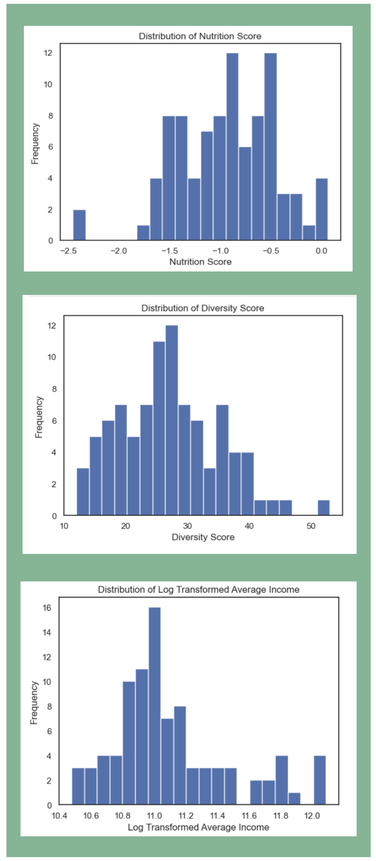
- A NutriScore is a food rating system that gives positive marks for a food's fiber or protein content and negative marks for a food's sugar and sodium content for example. NutriScores from –15 (very good) to 40 (very bad).
- The NutriScore is calculated by subtracting positive marks from negative marks. A NutriScore was calculated using the NutriScore Python package for each entrée among all 98 schools.
- Each entrée's fiber, saturated fat, sodium, and sugar was normalized for 100 calories before feeding it to the NutriScore calculation.
- Each school was given two NutriScores. One represented the average of all entrée NutriScores, and one represented the average of fruit, vegetable, and entree NutriScore.

Calculating Diversity Score

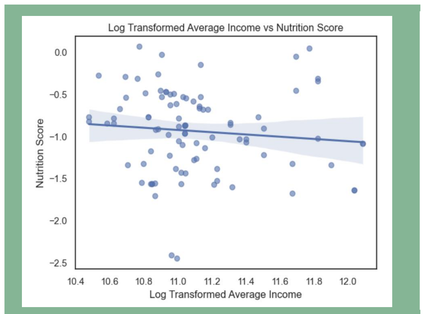
- For each school, unique items were counted in the categories fruit, vegetable, and entrée.
- Counts of entrees, fruits, and vegetables were averaged to create on overall diversity score for each school

District-level Averaged Income

- Financial data reflecting personal income for residents of the school district (as reported on the PA-40 income tax return) was downloaded from the Pennsylvania Department of Education government website.
- Calculated the average income by dividing the District level total adjusted income for 2025 by the amount of tax records submitted in that district.
- Log transformation was applied to the average income to normalize the right skewed distribution.

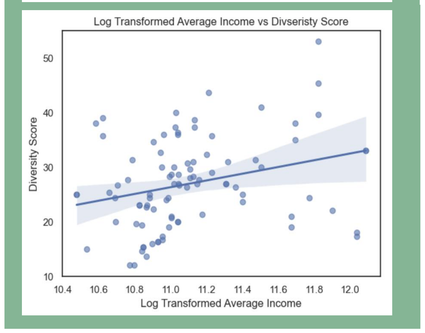


Findings

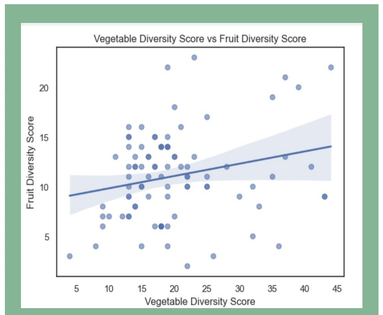


- Lower nutrition score corresponds to better nutrition.
- The regression suggests a slight negative association which means that nutrition improves slightly as average income increases, however this relationship is not statistical significance (p = 0.34).
- Income explains about 1% of the variation in nutrition scores (R^2 = 0.01).
- 10% increase in income is associated with about a 0.013-point decrease in nutrition score.

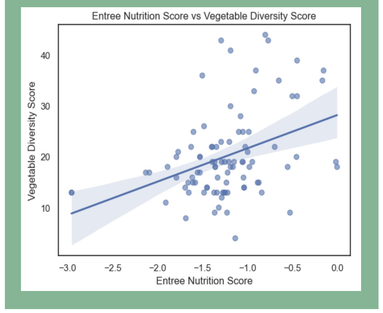
Log income	Dollars
10.4	$32,900
10.5	$36,300
10.6	$40,200
10.7	$44,400
10.8	$49,000
10.9	$54,000
11	$59,900
11.1	$66,200
11.2	$73,700
11.3	$81,400
11.4	$90,000
11.5	$99,500
11.6	$109,600
11.7	$120,900
11.8	$133,600
11.9	$147,800
12	$162,800



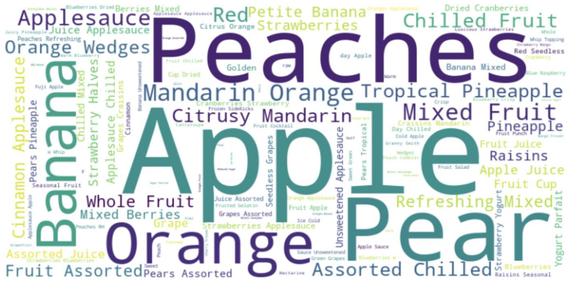
- Higher diversity score corresponds to greater diversity.
- There is a weak positive relationship between district-level income and lunch diversity scores with statistical significance (p = 0.0061)
- Income only explains 8.1% of the variation in diversity (R^2 = 0.081).
- These results suggest that while higher-income districts may offer slightly more diverse menus, income alone is not a strong predictor of menu diversity.
- 10% increase in income (roughly $5,000–$10,000 depending on the district) is associated with about a 0.62-point increase in diversity score.



- The regression suggests a slight positive correlation among fruit and vegetable diversity scores, and it is marginally statistically significant (p = 0.02).
- This shows that schools with more vegetable diversity generally have more fruit diversity.
- Vegetable diversity only explains about 5.7% of the variation in fruit diversity (R^2 = 0.057), so is not a strong predictor.



- The regression suggests a small positive correlation between entrée nutrition scores and vegetable diversity score, and it is statistically significant (p = 0.00017).
- This means that schools with less healthy entrée options generally have more vegetable diversity.
- One prediction for this is that schools with less healthy entrée options may also be categorizing french fries and high fat/sodium salads as vegetables.
- Entree nutrition scores only explains about 14% of the variation in vegetable diversity (R^2 = 0.14).



Discussion

The genesis of this project stems from the recognition of the critical role that school meals play in a student's diet and a desire to examine whether socioeconomic factors contribute to differences in lunch quality. Data collection presented an initial challenge, as there is no standardized method for reporting school lunch menus across districts. Menus are published in a variety of formats, including PDFs, district websites, and third-party platforms. Among these, Nutrislice appeared most consistently across districts, and was therefore used as the primary source to obtain more standardized and comparable menu data. To our knowledge, Nutrislice has not been used as a structured data source in prior research, and this study represents an initial effort to extract and analyze its menu data. Webscraping Nutrislice proved to be quite intensive, and our initial data analysis revealed that certain schools varied in their categorization, and a few schools didn't include categories at all (milk was categorized differently as beverage or dairy, and snack items such as PB&J uncrustables were categorized as entrees). Correct categorization of food items is important for our project since different categories such as condiments and snacks would score differently in the Nutriscore and their value is less important than more nutritious menu items that dominate a student's diet. Eventually we decided that recategorizing foods would interfere with the integrity of our data and that retaining the true categories would better reflect the food items that schools offer. Considering the inconsistent categorization, calculations for the nutriscore and the diversity score was only conducted on entrees, vegetables, and fruit. Limiting our analysis to only include categorized entrees, vegetables and fruit still introduces bias since some schools might not be categorizing the entirety of their menu which omits important food items from being considered in their scores. Future analysis should fill the missing categories to ensure that major food items are included in the diversity and nutrition assessment of all schools. Our findings indicate that income does not contribute to nutrition and menu diversity as much as we had anticipated. However, our scoring for nutrition scores seemed to be unreliable. Once the vegetable, fruit, and entree NutriScores were averaged for each school, they ranged from only 2.5 points. Among all the vegetables, fruit, and entrées, the lowest NutriScore was Tossed Salad, and it scored a –10. The worst NutriScore was Pico de Gallo with 5 calories and 55 mg of sodium, and it scored a 14. These results show that the NutriScore may be an inconsistent or unreliable scale for deciding which foods are nutritious. If a similar project was conducted, a different scoring process for nutrition should be considered which may include unique scoring parameters for different categories. Our diversity score proved to be more statistically significant revealing a weak positive relationship with log average income. This being said, missing categories prevent our predictions from being fully accurate and it only considers entrees, vegetables, and fruit while omitting categories such as milk, grain, condiments and beverage. Together, these findings highlight the importance of improving data consistency and developing more reliable metrics to better assess school meal quality across districts. Addressing these limitations will strengthen future analyses and provide a clearer understanding of how socioeconomic factors may influence the nutrition and diversity of school lunches.

Bibliography

- Cullen, K. W., & Chen, T.-A. (2017). The contribution of the USDA school breakfast and lunch program meals to student daily dietary intake. *Preventive Medicine Reports*, *5*, 82–85. https://doi.org/10.1016/j.pmedr.2016.11.016
- *Financial Data Elements*. (2025). Pa.gov. https://www.pa.gov/agencies/education/programs-and-services/schools/grants-and-funding/school-finances/financial-data/financial-data-elements#accordion-64c212361d-item-8bc6d944bc
- Kinderknecht, K., Harris, C., & Jones-Smith, J. (2020). Association of the Healthy, Hunger-Free Kids Act With Dietary Quality Among Children in the US National School Lunch Program. *JAMA*, *324*(4), 359. https://doi.org/10.1001/jama.2020.9517
- Studer, J. (2022). *pyNutriScore (Version 0.0.2)* [Computer software]. PyPI. https://pypi.org/project/pyNutriScore/